

hog — beware. In *UT3* we see the ZOTAC GTX 260² score equally at both resolutions, meaning that the 216 SPs are actually working wonders for NVIDIA, as the 192 SP versions of the GTX 260 both have a drop in fps at higher resolutions. Company of Heroes and S.T.A.L.K.E.R. are very playable — these cards will play most games at the highest available settings. Amongst the GTX 260s both the ZOTAC cards impress the most.

At identical prices of Rs 22,999 the ZOTAC GTX 260 AMP! Edition and the ZOTAC GTX 260² make very good buys for someone looking for a fast gaming solution that is likely to be future proof for around another year.

NVIDIA GeForce 9800 GX2

Fun x 2, ditto heat

The 9800GX 2 is a card that some of us feel should never have been born. Of course some of us feel the same way about the 9800GTX, which was wrongly touted as the new high-end part from NVIDIA, but that's history now. Instead NVIDIA should have launched the GTX 280, which is nearly as fast as the 9800 GX2 at times, and faster than it most of the time. Like the 7900 GX2, the 9800 GX2 is a double PCB card sheathed in a metal casing that also acts as a passive heatsink. Unfortunately the 9800 GX 2 is a very hot card, and the fact that both PCBs are less than an inch apart, ditto the 128 SPs that each of the two 9800 GTX cores sport. Sporting a whopping 256 SPs the 9800 GX2 is a powerful solution. It also gets 1 GB of RAM, 512 MB for each GPU. In terms of pricing the 9800 GX2 has experienced a 25 per cent drop in its original pricing. Despite this, it continues to be a slow seller for NVIDIA — the reason, it's a dual GPU solution that not everybody may want to use. The plus — you can enable SLI on this card without needing an SLI ready motherboard.

Features

We tested just three cards based

on this chipset — two from ASUS, and a single card from XFX. While the other two were clocked at the NVIDIA-recommended 600 MHz the ASUS EN9800GX2 TOP was clocked at 670 MHz. This card ships in a huge box that includes a neat leather CD case and mousepad — something that ASUS ships with all its cards. Also bundled was a full version of *Company Of Heroes: Opposing Fronts*, which is a superb game. The plain jane ASUS EN9800GX2 also ships with the same game. We feel the choice of bundled game is as important as just giving a new game that nobody would want to play. All these cards sport the reference NVIDIA cooling assemblies (which we feel is woefully inadequate). Also integrated is an HDMI port, apart from the two regular DVI ports.

Performance

The 9800 GX2 is a very solid performer. Although it doesn't run away with all the accolades we'd like to point out the consistency of the results, which is something NVIDIA is to be commended for. This trio is the only one to consistently offer more than 30 fps on *Crysis*. *S.T.A.L.K.E.R.* still favours the more powerful GTX 280 at lower resolutions in particular. *Company Of Heroes* shows it is severe with multi GPU solutions and the 9800 GX2s get murdered by every other high-end card in this comparison — NVIDIA — fix your SLI drivers. All these cards are able to chew up *F.E.A.R.* and *World In Conflict* gives a consistent 35 fps — which doesn't say much other than our CPU being a bottleneck. The over-clocked ASUS EN9800GX2 TOP is the only card to crack the 11,000 mark in 3D Mark Vantage — which proves its DX10 shading power. The 9800 GX2s are faster than the GTX 260s and ATI's HD 4870 cards and HD 3870 x2 cards. Only the Radeon HD 4870 x2s and the GTX 280s are faster. This being said, we feel that such dual GPU solutions are best avoided until application support is up to par.

We do not recommend investing in a 9800 GX2 based card now. There are two reasons for this. One, the GTX 280 is a much newer architecture, and is at par in terms with performance. Secondly this card generates a lot of heat, and the cooling needs special attention. Moreover multi GPU solutions don't always scale linearly in performance.

ATI Radeon HD 4870 x2

Born out of a indomitable obsession with being the fastest

Concieved out of two Radeon HD4870 GPU cores on a single PCB the HD 4870 x2 is a monster to be sure. It features 1600 SPs (800 from each 4870), and features an unheard of 2 GB of GDDR5 memory that operates at a whopping 3600 MHz. The phrase *gross overkill* comes to mind. AMD has gone all out with this one, and the 4870 x2 is supposed to do what the single GPU version couldn't — dethrone the mighty GTX 280, and restore ATI's place at the helm of the 3D performance realm — a position it hasn't properly regained after its now ancient 9800 Pro cards. This card looks very similar to the earlier Radeon HD3870 x2, but it's slightly lighter, and a hell-of-a-lot lot faster. The 4870 x2 is a very hot card, a little hotter than NVIDIAs 9800 GX2s and the power requirements of this card are astronomical — the card will consume close to 300 watts while on full load.

Features

The only 4870 x2 we got was from ATI itself. This card is also available from ATI vendors like GeCube, ASUS, Sapphire and VisionTek. Each of the two cores is clocked at a super fast 750 MHz, and the memory runs at a superlative 3,600 MHz. The card itself has no HDMI port, and ATI chose not to provide an adapter — but this was a review sample only. The card itself looks very long and sleek, and is built well. The heatsink extends to the rear side of the card. This aids dissipation in our opinion.

Performance

The 4870 x2 is very fast, but the usual issues with CrossFire support throughout applications resurfaces. It demolishes all the GTX 280s in *Crysis*, and manages an excellent 31 fps with all DX10 effects set to maximum. *S.T.A.L.K.E.R.* is another benchmark that is a runaway ATI victory with this card. *Company Of Heroes* doesn't scale with multi GPU solutions (as usual) and the 4870 x2 gets a blemish on its (until now) spotless scorecard. If it's any consolation it outpaces NVIDIAs dual GPU offerings by a wide margin, but cannot match the GTX 260 / 280 duo. *F.E.A.R.* also doesn't do much for CrossFire solutions and this card languishes behind even its single-GPU stable mates.

We feel the 4870 x2 has a lot of potential, but there are a couple of issues. The first is application support — more game

developers need to add proper CrossFire support in their games. This card is a mixed bag. It's very fast at times, and abysmally slow at other times — ATI needs more consistency here. We'd also like to point out the heating issues — this is a hot card that will likely fry in our summers. If only ATI could get 1600 SPs on a single GPU core...

ATI Radeon HD 4870

The giant killer

ATI's new HD4870 could well be a poster child for a comeback advertisement. Let's face it — its GPU fortunes have been dwindling since AMD took it over. NVIDIA enjoyed winning products across the high-end and mid-range spectrums, as well as some great entry level offerings. Enter the core AMD labelled the RV770 — which has spawned two GPU variants that have radical-

ly changed what we expect to pay for a fast graphics card. The HD4870 is a high-end part that is priced as a mid-range offering. It consists of an 800 SP based GPU — which is a huge hike considering its 3870 had just 320 SPs. Although a direct comparison between ATI's 800 SPs and NVIDIAs 240 SPs is impossible due to differences between their SP architectures, the HD4870 is a step in the right direction for ATI. The best thing is it doesn't cost you much and here's why — it's built on a cost saving 55nm process, ATI also simplified the PCB by sticking to a 256-bit memory interface. What it did was to slap on shiny new GDDR5 memory, which itself is a very new, and as yet untested standard, which pioneers like Samsung and Qimonda are pushing. Therefore ATI can cut costs much more effectively than NVIDIA can with its GTX 2xx architecture.

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Radeon HD 4870 SONIC Dual Edition

Dual Fans

Dual BIOS

Ultimate Dual Output

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Features:

- GDDR5 512MB 256bit
- Overclocking Edition
- HDMI + DisplayPort + DVI + D-SUB
- Triple-Pipe cooler with dual fans
- Smart BIOS switch

Other features shown: Nickel-plated copper pipes, RADEON HD 4870, PALIT logo.